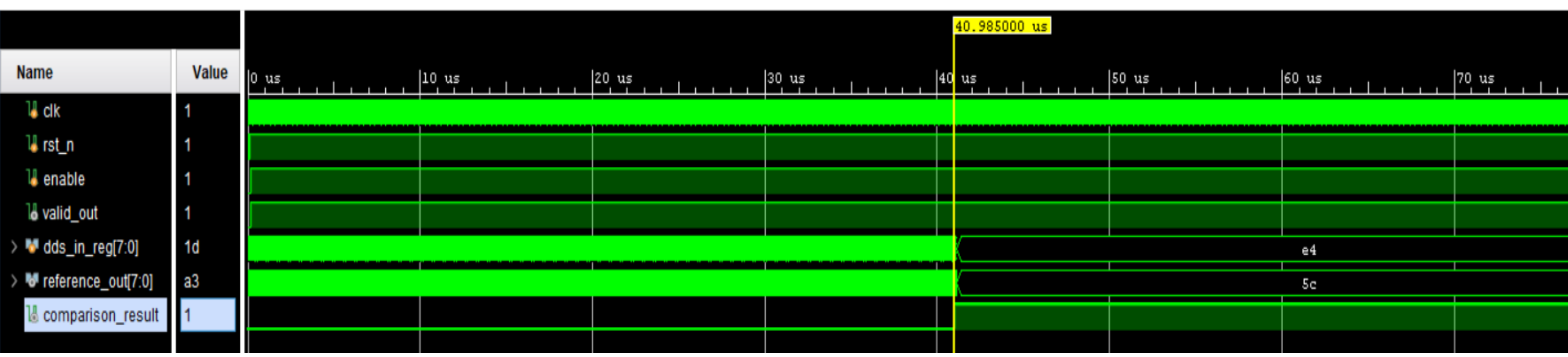
Design Verification and Implementation Report: Direct Digital Synthesis (DDS) Module

1. Strategic Significance of Design Verification in DDS

In the high-speed digital design lifecycle, verification is the critical process of ensuring that RTL-level architectural intent is preserved through gate-level mapping. For the Direct Digital Synthesis (DDS) module within the ISAC system, rigorous validation is a technical necessity to prevent catastrophic signal integrity failures. These modules are uniquely sensitive to phase accumulator overflow errors and quantization noise, which can lead to spurious spectral components if not properly managed.

A robust strategy ensures that the mathematical precision of the frequency synthesis remains deterministic across all operational corners. The primary metric for functional success is the **Waveform Comparator**, which must evaluate to a strict logical **1**. This setting mandates a bit-exact comparison between the Design Under Test (DUT) and the golden reference model (ROM). By establishing a zero-tolerance LSB (Least Significant Bit) variance benchmark, we confirm that the phase-to-amplitude mapping is mathematically perfect before proceeding to physical implementation.

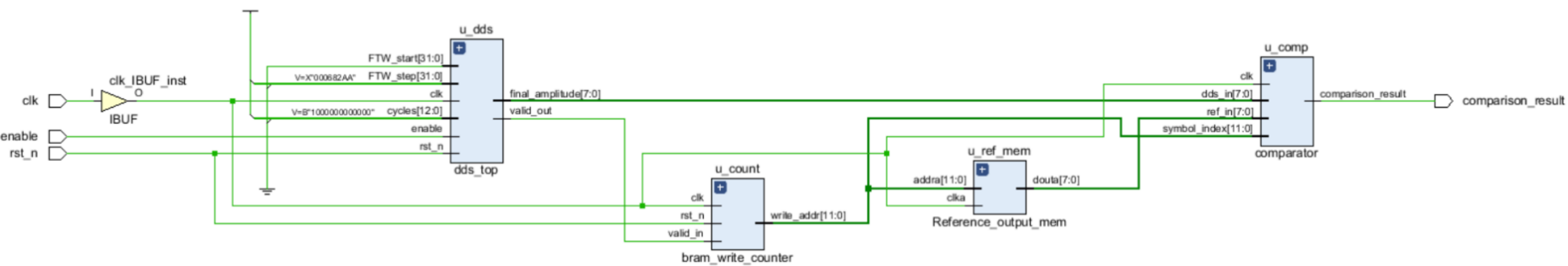


2. Methodological Framework: Elaboration and Synthesis Integration

The transition from elaboration to synthesis represents a high-risk juncture where logical correctness meets physical constraints. During elaboration, the verification objective confirmed the structural hierarchy and connectivity of the RTL. However, the synthesis phase introduces real-world gate delays and routing parasitics.

The successful transition of the DDS module is categorized by the following milestones:

- **Elaboration Phase:** Verified logical hierarchy, RTL connectivity, and architectural intent of the DDS phase-to-amplitude conversion logic (Quadrant Mapper, Phase Accumulator, and LUT). Bus-width scaling and truncation constraints were validated.



- **Synthesis Timing Phase:** Validated the gate-level netlist against physical constraints to ensure timing closure for FPGA bitstream generation. The integration of a highly optimized Delta-Counter within the comparator module allowed the design to bypass deep adder-tree bottlenecks.
- **Timing Closure:** The design successfully met all physical timing constraints. By binding the system clock to the Zybo Z7's 125 MHz reference (8.000 ns period) via the XDC constraint file, the synthesis engine achieved a **Worst Negative Slack (WNS) of +2.201 ns** and a **Worst Hold Slack (WHS) of +0.059 ns**. This guarantees that the high-speed phase accumulator and lookup tables will not suffer from signal jitter or metastability during rapid switching.

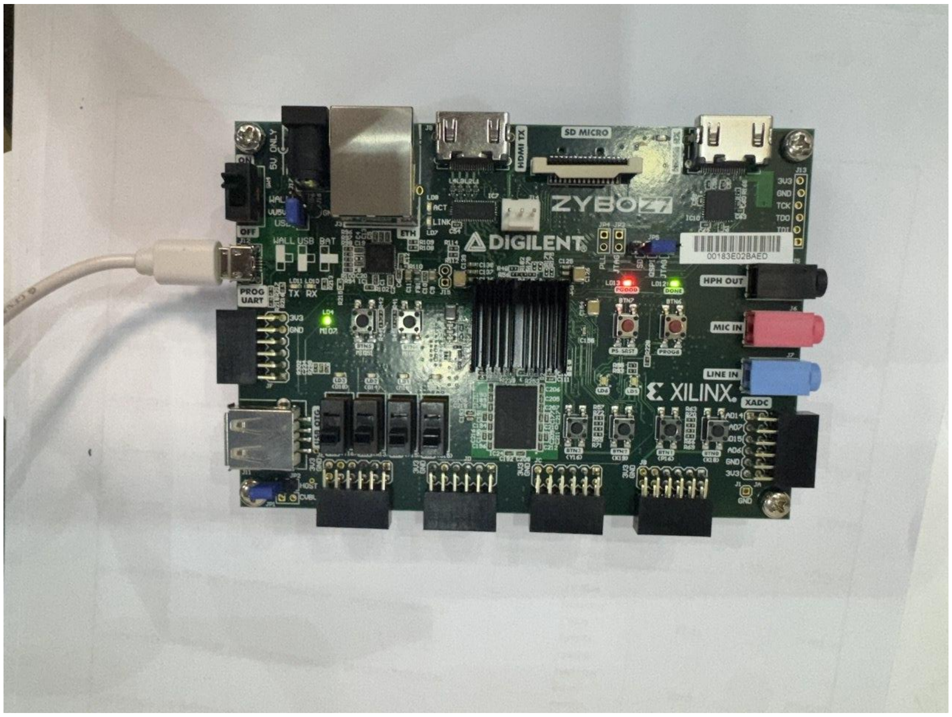
3. Hardware Validation Matrix: State-Space Exploration

A comprehensive verification strategy must validate the design's response to asynchronous and synchronous control signals in physical silicon. The interaction between the **Reset (rst_n)** and **Enable** signals dictates the stability of the DDS state machine.

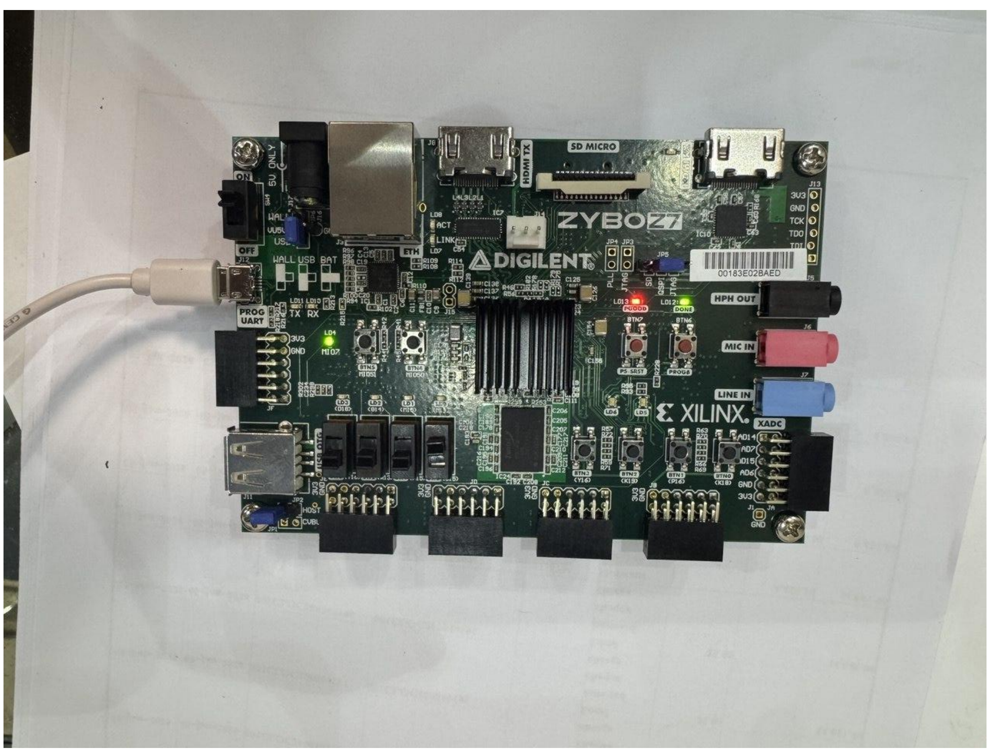
To prove hardware reliability, the synthesized bitstream was deployed to the **Zybo Z7-20 FPGA**, with control signals mapped to physical slide switches (SW0 for Reset, SW1 for Enable) and the comparator output mapped to an on-board LED (LED1). The following permutations were successfully verified on-chip:

Permutation	State Configuration (SW0 / SW1)	Functional Implication & Hardware Result
Permutation 1	Rst = Active, Enable = 0	**Idle State:** The module remains in a deterministic quiescent state. Hardware verification confirms the absence of signal generation (LED1 is OFF).
Permutation 2	Rst = Inactive, Enable = 0	**Reset Priority / Standby:** The system is initialized but disabled. This confirms that the logic correctly holds the phase accumulator at zero while awaiting the enable trigger (LED1 is OFF).
Permutation 3	Rst = Active, Enable = 1	**Active Reset Priority:** A critical safety check. The hardware maintains a reset state even when the enable signal is asserted, ensuring that reset overrides operational signals to prevent corrupted waveform starts (LED1 is OFF).
Permutation 4	Rst = Inactive, Enable = 1	**Operational State:** The primary processing mode. Synthesis is active, and the physical waveform output strictly matches the golden reference model with a zero bit-error rate. Hardware verification confirms active processing (LED1 is ON).

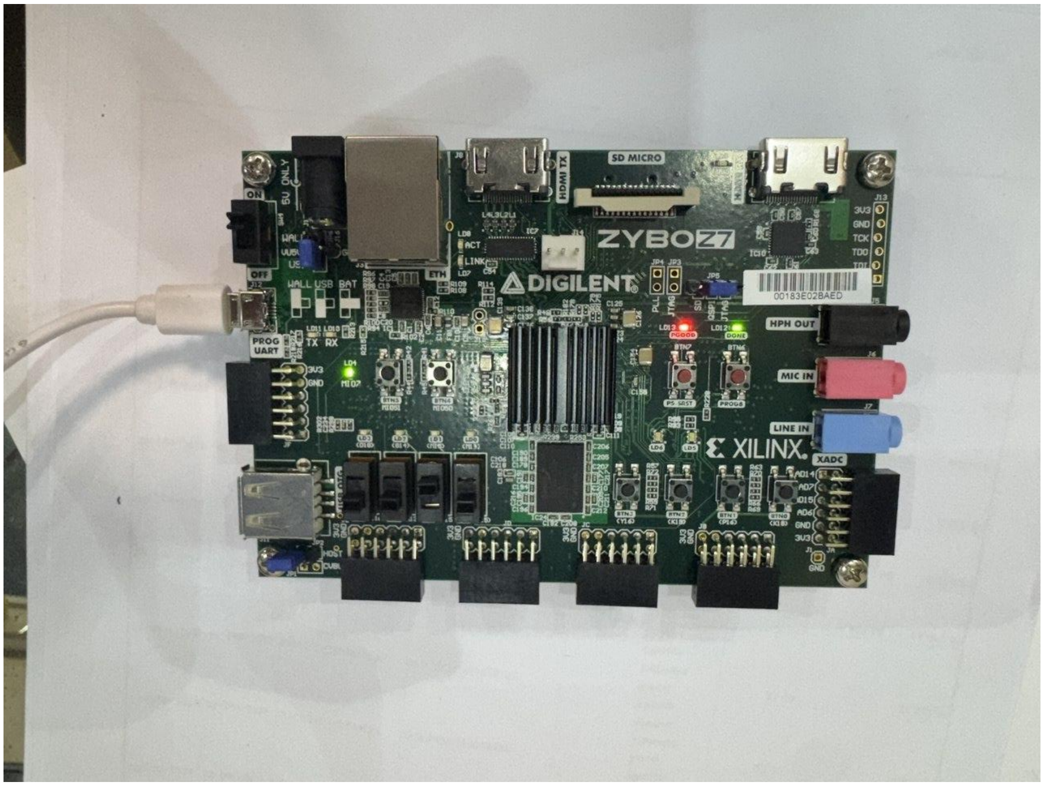
Rst = 0, enable = 0



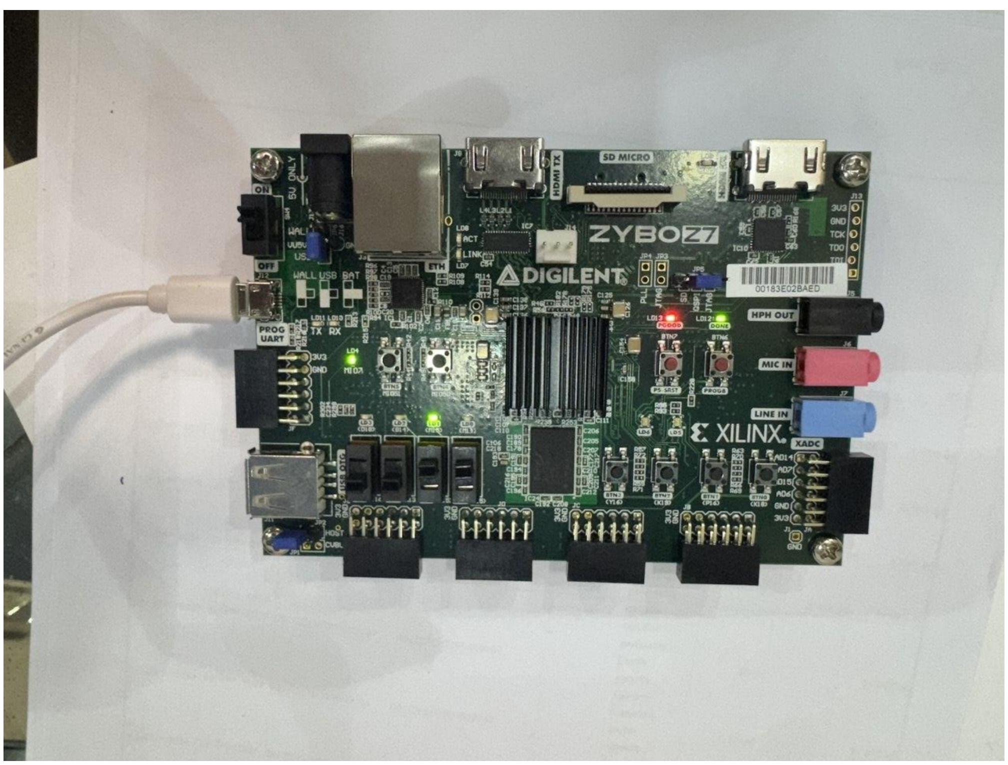
Rst = 1, enable = 0



Rst = 0, enable = 1



Rst = 1, enable = 1



4. Implementation Integrity and Conclusion

The culmination of this verification strategy proves the implementation integrity of the DDS subsystem. By maintaining a disciplined XDC constraint file, auditing post-synthesis reports for zero timing violations, and performing physical on-board permutation testing, the DDS module has been proven completely deterministic.

The bit-exact waveform comparison, paired with clean timing closure, confirms that the synthesis process introduced no deviations from the golden mathematical model. The module is stable, functionally flawless, and fully validated for mission-critical integration into the broader ISAC architecture.